Find the directrory in `/root/86344/`, where the onwer is not the `root` user. Write the full (absolute) path name to the file: `/root/86344.ans`

```
[root@exam-vm ~]# find /root/86344/ -not -user root >/
root/86344.ans
```

Create a new user with name `user86315` and lock the password of the user!

```
[root@exam-vm ~]# useradd user86315
[root@exam-vm ~]# passwd -l user86315
```

Who signed the file `/root/contract-86302.txt`, *Anne*, *Bob*, *Carol*, *Daniel* or *Eve*?

The signature was created with the help of the SHA 256 hash algorithm and stored in the file: `/root/contract-86302.sha256.signature`

- The public key of *Anne* is in `/root/Anne-86302.pub.pem`
- the public key of *Bob* is in `/root/Bob-86302.pub.pem`
- the public key of *Carol* is in `/root/Carol-86302.pub.pem`
- the public key of *Daniel* is in `/root/Daniel-86302.pub.pem`
- the public key of *Eve* is in `/root/Eve-86302.pub.pem`

Write the answer (the name) to the file: `/root/86302.ans`

```
[root@exam-vm ~]# openssl dgst -verify /root/Anne-8630
2.pub.pem -signature /root/contract-86302.sha256.signa
ture /root/contract-86302.txt
```

```
[root@exam-vm ~]# echo Carol > /root/86302.ans
```

Download the file `5d3a8e1b` with the help of the command below.

```
curl -o /5d3a8e1b https://progcont.hu/
```

The file contains an encrypted filesystem in *LUKS* format. The password is: `5d3a8e1b`. Copy the content of the filesystem to the directory: `/root`

```
[root@exam-vm ~]# losetup -f /5d3a8e1b
[root@exam-vm ~]# cryptsetup luksOpen /dev/loop0 loop
Enter passphrase for /5d3a8e1b:
[root@exam-vm ~]# mnt /dev/mapper/loop /mnt
-bash: mnt: command not found
[root@exam-vm ~]# mount /dev/mapper/loop /mnt
[root@exam-vm ~]# ls /mnt
070f  lost+found
[root@exam-vm ~]# cp /mnt/070f /root
[root@exam-vm ~]# cp -r /mnt/lost+found/ /root
```

Decrypt the `/root/86261.enc` file by using `86261` as the key and 0 as the initialization vector. The file was encrypted by using the `aes-128-cbc` algorithm. Save the result as `/root/86261.txt`.

```
[root@exam-vm ~]# openssl aes-128-cbc -d -in /root/862
61.enc -K 86261 -iv 0 > /root/86261.txt
```

Decrypt the `/root/86257.enc` file by using the `86257` password and the `des-ede` algorithm. Save the result as `/root/86257.txt`.

```
[root@exam-vm ~]# openssl des-ede -d -in /root/86257.e
nc -k 86257  > /root/86257.txt
```

Sign the `/root/86247.txt` file with the following private key. Use the SHA 256 hash function. Save the signature as `/root/86247.sign`!


```
-----BEGIN PRIVATE KEY-----
MIICdgIBADANBgkqhkiG9w0BAQEFAASCAmAwggJcAgEAAoGBALGxXdQslY2y33Hr
6JBmdQL4fJmDZolEKJIojg9IapEI95SV5gsZb9+K6J+1sJAeq0k4wzm6LNWdMT3n
xDaWM0rN/wvM+MzMDoz2IOep6I/VQMFvq9/kL0iwhM5w6NJtkIfwNJhimw5KsEF7
iAtr9xl+3Gx6VIZ5ZgzpSHOeFspTAgMBAAECgYBpfApgX0s0wCAHZ+06c6g46DxH
gYrIJ+8RvQOALRQeHz2iNk5G/oW8JaEs1lYHaD10jT7PsSbWdKd7wW9onEe7vcfx
miJbcZaXON6pY11BZOVaGgSMWR2/frQVVPH9pdVk5L+UyJfE/kNGgdgAnKU8+jRT
yeMUer8bh5/GSJL8YQJBAONwyYVuOQo2dkVZTtGormh28NDt3o8NK9NaKUlzjS3E
YvzPdABDhjY+kjeA0LhT1xVwFr7MgekGks/EgbKlbbECQQDIAWifEguMgV0fCYB/
gXJTNNqbNgLf3+tQBNru8HCo1Ed4HPscx5tpfZHieXUW63PYbZC/kofMmoZ1e+Dc
mKVDAkBiEuyTIOhvwvRVCyG1vqsWWNOXBDuILAHN7X8IMU+bgKe+pCY9RuDE205p
qJ5YHa8Ni3wDYmRSe4crGG1k/3jBAkEAss3/mGZriLuGYt790AAMEzMnVKdevoLT
PoB4sjCmp2jQVCOko8AXwqGOMKhg85KfyJd7VqlLWrGzD4kmKFEXPQJAEj/QSzkJ
Dmqu2Q+KrIyYSDfWwdFDda6qkH9QHoTQ2aE6RtkShhAShA6EsJdOX2d3tCuSdZ2B
2o0qC6Dvgv2y/A==
-----END PRIVATE KEY-----
```


```
[root@exam-vm ~]# umask 077
[root@exam-vm ~]# cat >k
```

Kulcs bele

```
[root@exam-vm ~]# openssl dgst -sign k -sha256 -out /r
oot/86247.sign /root/86247.txt
```

Set the permissions of the `/root/86237.txt` file to read-only.

```
[root@exam-vm ~]# chmod 444 /root/86237.txt
```

Give read-only permissions to the `operator` user on the `/root/86224.txt` file with a new USER ACL entry. (Do not change anything else about the file and do not give or take any other permissions.)

```
[root@exam-vm ~]# setfacl -m u:operator:r /root/86224
txt
[root@exam-vm ~]# getfacl /root/86224.txt
```

Set owner of the `/root/86211.txt` file to the `operator` user. (Do not change anything else about the file.)

```
[root@exam-vm ~]# chown operator /root/86211.txt
```

Create a file with name `/root/86198.dat` and with a size of 861980 bytes.

```
[root@exam-vm ~]# dd if=/dev/zero of=/root/86198.dat b
s=861980 count=1
```

Generate an `1024` bit RSA key and save it as `/root/86191.pem` as a `PKCS8` private key.

```
[root@exam-vm ~]# openssl genpkey -algorithm RSA -pkey
opt rsa_keygen_bits:1024 -out /root/86191.pem
```

Create a new `/dev/sdc1` partition on the drive `/dev/sdc`. Format the partition with `ext2` filesystem and mount it to the `/mnt` folder in read only mode.

```
[root@exam-vm ~]# fdisk /dev/sdc
```

N

Command: w

```
[root@exam-vm ~]# mkfs.ext2 /dev/sdc1
```

```
[root@exam-vm ~]# mount -o ro /dev/sdc1 /mnt
```

Allow the owner of the following private key to log in as the root user.


```
-----BEGIN PRIVATE KEY-----
MIICdwIBADANBgkqhkiG9w0BAQEFAASCAmEwggJdA
GPNsk+p9HFUTwSM1ka80tXOHu8O+bqXEf86g+gMOB
v4ijljj/6xJcCqPP2LVeYVM1aDWjYADHmHRFOJwLC
u7RSsX9W8mCAqx1xlpfkpiQiiTu5AgMBAAECgYBjE
EOVlh/BTl+UY65NXFg7jneCo2zdnPKVJwtlodtpZ+
D70FVscj9TSIzP1lHBNcWdt1qkZPR1ADnauueg8l8
H0cY6qRQjkNgipJVCQJBAO2RAVk21E34jE6+6w2/j
2DDkBxVHKYLdtnnL9eX3O69T4UKYEZjWq58GkGXqA
kfTW7x4pFwgjLEyd+Kzs5LpmU8MxEL6ndXZVHnmy1
z4qjAkAF1R6MWy1BMM35DPTmZIKG9JA/+1kyPKsZc
+ary12wOtOd95A/iibqy781RwzoDAkEAhAS4P8Ow5
b7qMv9aETRajXJ7AXVdnff1rX9uyVFydlW3yMdMxm
tTaU838GJB5L8Afq/HIW+Biw/GmgcuKMhQPRE0dIk
aEUNnE4+0e7tQXE=
-----END PRIVATE KEY-----
```


```
[root@exam-vm ~]# cat >kk
```

Kulcs bele

```
[root@exam-vm ~]# stat .ssh
stat: cannot stat '.ssh': No such file or directory
[root@exam-vm ~]# umask
0077
[root@exam-vm ~]# mkdir .ssh
[root@exam-vm ~]# ssh-keygen -y -f kk >.ssh/authorized
_keys
[root@exam-vm ~]# stat .ssh
```

Create the user160184 user and generate a new default ssh RSA key (/home/user160184/.ssh/id_rsa) for it without passphrase. Allows user160184 to log as root with the help of the RSA private key.

```
[root@exam-vm ~]# useradd user160184
[root@exam-vm ~]# su user160184
[user160184@exam-vm root]$ ssh-keygen
```

```
[user160184@exam-vm root]$ ssh-copy-id root@localhost
```

Find the directory in /root/160171/, in which the sticky bit is set. Write the full (absolute) path name to the file: /root/160171.ans

```
[root@exam-vm ~]# find /root/160171/ -perm /1000 >/root/160171.ans
```

Configure the user account operator so that its password expires on 2021-07-24 and its maximum number of days between password change is 100.

```
[root@exam-vm ~]# chage operator -d 2021-04-01

[root@exam-vm ~]# chage operator -M 100
```

Set the permissions of the file /root/160101.txt so that its permissions shows as -r--rws--T in the directory listing.

```
[root@exam-vm ~]# chmod 3470 /root/160101.txt
```

Delete all ACL entries of the file /root/160093.txt!

```
[root@exam-vm ~]# setfacl -b /root/160093.txt
```

Set ownership of the file /root/160076.txt so that the owner will be operator and the owner group will be users. (Do not change anything else about the file.)

```
[root@exam-vm ~]# chown operator:users /root/160076.txt
```

Copy the first 70 byte of the file /root/160067.dat to the file /root/160067.part.

```
[root@exam-vm ~]# dd if=/root/160067.dat of=/root/160067.part bs=70 count=1
```

Create a new `crypto_LUKS` encrypted `xfs` filesystem on `/dev/sdc` and mount it to the `/mnt` directory.

```
.............
[root@exam-vm ~]# cryptsetup luksFormat /dev/sdc
```

Yes

Jelszó pl APPLEapple

Jelszó újra

```
[root@exam-vm ~]# cryptsetup luksOpen /dev/sdc x
```

APPLEapple

```
[root@exam-vm ~]# mkfs.xfs /dev/mapper/x
```

```
[root@exam-vm ~]# umount /mnt/
[root@exam-vm ~]# mount /dev/mapper/x /mnt
```

Dekódolja az alábbi privát kulcs segítségével kódolt `/root/287304.enc` állományt. Az eredmény a `/root/287304.txt` állományba kerüljön!


```
-----BEGIN RSA PRIVATE KEY-----
MIIEpAIBAAKCAQEAuI4w+hLD5XXDfdWhStYJMxWn13QNgbf9nSy8P5Tl8aEp7RA4
9FANsyT6E1yFyxcYh0DLvgx47dNp+xHHD7h3a7Gv+hG+ipPbem6+3WoBOsWg+Zz9
0D5Jeu7TVMZqYdmH6ngdaG5qj4R4XZ6UpjhH34TXlyE1n0XlZB/27qDgEDkwT8wd
FNI2a//SvyLviqrGQ3vZB+1TsCMIJSsc9UFHtbl3O008pVqsEPMgTgT1cW++GS6n
zKR5p2mnR9D9mpKimy2uWXjvmWUWamqXanw2rF5ghlwxgnsR8zjQyuuo3S5KLBU9
ZTjJVDNrEwJtYaHIyTOjESgp6ptZ4zUIQ8r/UwIDAQABAoIBAQC0J2k707GSfiB0
52LpUIHucgdDVDATHm3dTdv9HPZRpobsuNc/9h1JcKEE4V5hX4w8C05XmIk7bxKX
WUU93U4FvJARxdyAHdEzdIOwU1gb7Os0fsFs1ECaViPJL16NYfFBOjtKw+4VZrSw
fqqQj4Ppdx2Uk4Bql6pS0LkVO8k9S0qmM0MfT7LWigd0uc25eppXAFNkB1TWwW28
rGDxaZndFkO4RvfLL9VXA2uYM3ObgyaPrtaYfhCJ30lHtK3HVn6B6/QWu/djxEXl
dhosc7tygnmCinsa23Mc6xiqC4PXs/0soKBv7gSginLzBnvIg6tFYhzXNwdhJ/KY
zlQoE+1BAoGBAPUa5HHxZ5QSm9ajf8zyAembby99D8z5oli2evtHuhV5IqZPSPE0
evLD5r/fTEdeL42vJH5CLfxKn93djyIvNQ9kPa4/2jlzMesAkyMuu2RcljAfn1J9
b34LuZZfgGdE0/GKi6No0UUMlD8tKcSUqdL4+iJdyIlZSLtUGyLw4VNxAoGBAMDC
Sm5sYwvnMltNuXyGEyNJZn08Kp454gr5ILb+zpSDi6HgbOGCXtp2issqVmKDPLkx
x3xu5679qYuJt9vueURVOE9KHIe3JerkbWugsMqagjdwq3tFE7eY6YdwNlK7tDYS
DEpTzlkn8Mppjt/JeS567a3VbYrgzDa4x0QxWtUDAoGAenbyV4AfmZtmo+rTSypM
DFWuSxL4lxjPpfWfIvgmoN4CkwggUOxA1hVNsOa4LRyBNBCjcjZM1bbPg8JU5uah
Y0xZiTu8JGJLObjyy2xpEu8z112ldEjaMSdYliqndHGuWAw6Uu5nSVHMrHg7W05Q
P+pIyG0FaqLitT6DxISIgPECgYBqb95hsv/OJs+rkvOFSxNuPK2XIY326Mdkf8QM
yA6R+0EzFZWJwE9YGnTxUmN46jaxnEcm8qaD/hWMs3MOwlO748LAJZAzcHJDjfbN
o2F7vm2A53oJC0+j5Y+t9cxJStRrtvOKZ1Rd7qwLWNZHxBxiCgcMIfnm4dJhtQ5z
HT2muQKBgQDL+ygG9RdHKYGXaFNENOr+oC8TU+8GDjYKzu8ZNbFTENq74DUOwP+n
3rfFKMuJHHxrrx/T5K3jDQKqYmxP4S7lqOV/pI8Y9MO++u1fIlQwoljoA62+K6Dl
XgvGBPWpfZM6jA8vrrPh1D8DHrPbCGQbIRJ3GdOJUvXR51L86pSw8A==
-----END RSA PRIVATE KEY-----
```


nano /root/private.key

Paste

openssl rsautl -decrypt -inkey /root/private.key -in /root/287304.enc -out /root/287304.txt